

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO1: *Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Ramamoorthy

QUESTIONS: 12

Total Marks:60

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Visahli.s(192521049) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

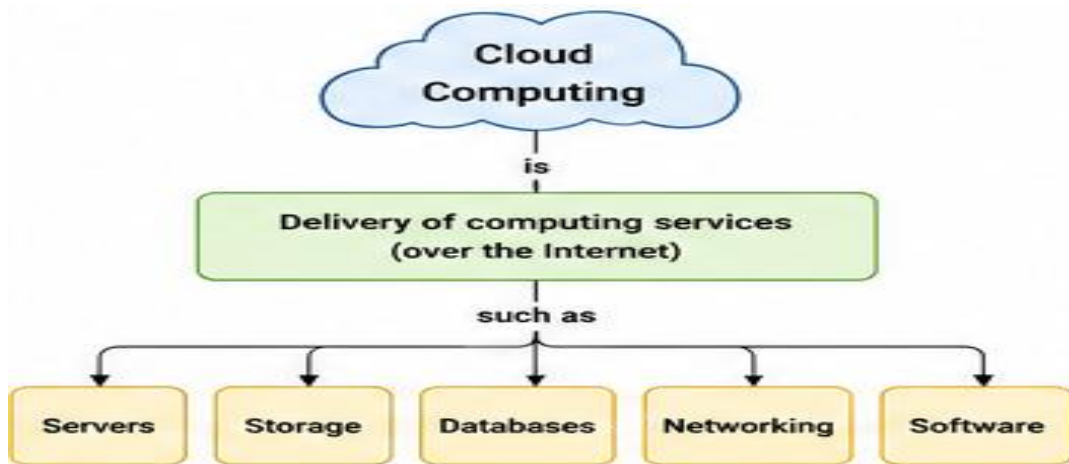
I understand that any violation of academic integrity rules will result in disciplinary action.

Vishali.S

Signature of the student:

CONCEPT MAPPING

Question 1



Explanation:

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, and software over the Internet. Instead of buying and maintaining physical hardware, users can access these resources on demand and pay only for what they use.

Characteristics

- **On-demand Self-Service:** Users can access resources whenever needed.
- **Scalability:** Resources can be increased or decreased based on demand.
- **Elasticity:** Resources automatically adjust according to workload.
- **Resource Pooling:** Multiple users share computing resources efficiently.
- **Measured Service:** Users pay only for the resources they consume.

Benefits

- Reduces infrastructure cost.
- Provides flexibility and accessibility.
- Offers high availability and reliability.
- Enables quick deployment of applications.
- Supports business growth through scalability.

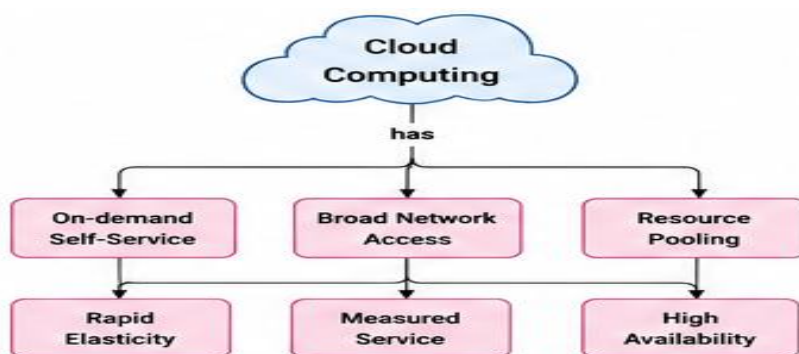
Limitations

- Security and privacy concerns.
- Requires a stable Internet connection.
- Vendor lock-in may occur.
- Service outages can affect applications.

Analysis

Cloud computing has transformed modern computing by providing scalable and cost-effective services over the Internet. Its characteristics, such as elasticity and resource pooling, allow organizations to optimize resource utilization while reducing operational costs. Although cloud computing offers many advantages, including flexibility and global accessibility, it also introduces challenges such as security risks and dependence on Internet connectivity. Organizations must balance these benefits and limitations when adopting cloud technologies.

QUESTION 2 :



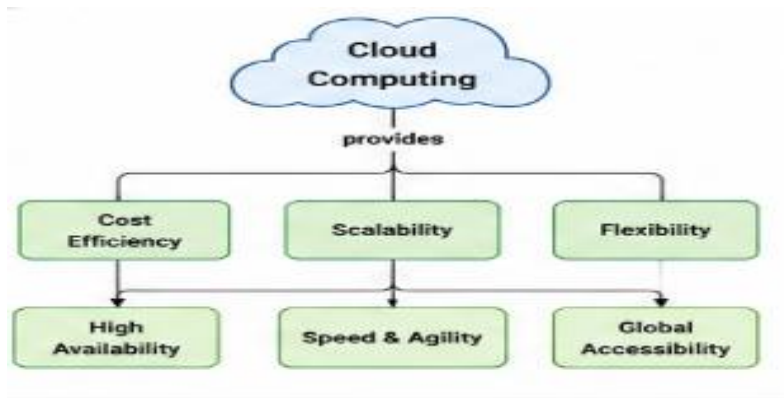
Explanation

Cloud computing evolved through several technological advancements. Initially, computing relied on standalone hardware and mainframe systems. As networking technologies improved, distributed systems enabled multiple computers to work together efficiently. The evolution of Internet software, from static web pages to dynamic web applications and web services, made online service delivery possible. These developments eventually led to cloud computing, where computing resources are delivered over the Internet on demand.

Analysis

The evolution of cloud computing demonstrates how advancements in hardware, networking, and Internet technologies transformed traditional computing into a flexible and scalable service model. Hardware virtualization improved resource utilization, distributed systems increased reliability and scalability, and Internet software enabled remote access to applications. Together, these technologies support modern cloud services by providing cost-effective, highly available, and on-demand computing resources.

QUESTION 3



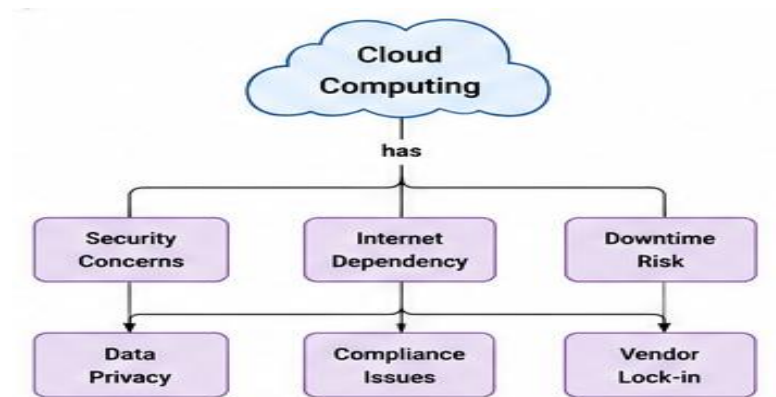
Explanation

Cloud service models define how computing resources are delivered to users. **Infrastructure as a Service (IaaS)** provides virtual machines, storage, and networking, where the cloud provider manages the infrastructure while the user manages the operating system and applications. **Platform as a Service (PaaS)** provides a platform for application development, with the provider managing the infrastructure and runtime environment, while the user develops and manages applications. **Software as a Service (SaaS)** delivers complete software applications over the Internet, where the provider manages everything and the user simply uses the application. **Anything as a Service (XaaS)** is a broad model that offers various cloud-based services such as storage, databases, and security through the Internet.

Analysis

Cloud service models offer different levels of management and responsibility. IaaS provides maximum control but requires more user management. PaaS simplifies application development by managing the platform. SaaS offers the easiest experience because the provider manages the complete software environment. XaaS extends cloud computing by delivering specialized services on demand. Choosing the appropriate service model depends on the project's requirements, budget, and the level of control needed by the user.

QUESTION 5



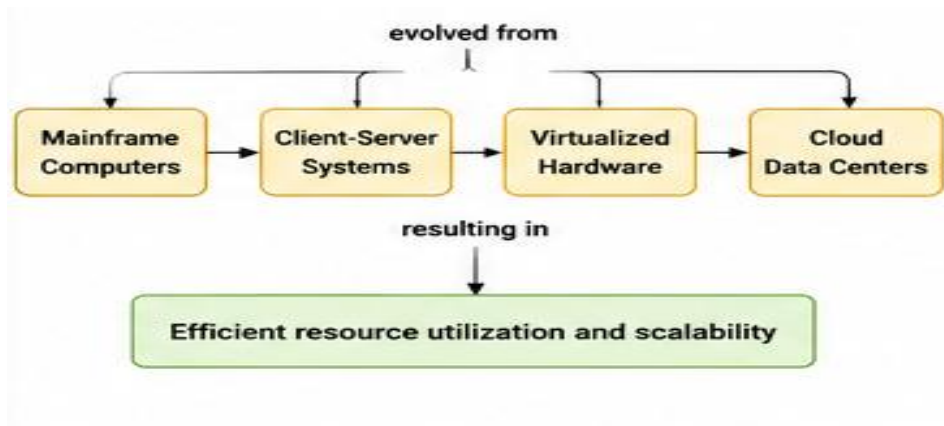
Explanation

Virtualization is the technology that allows multiple virtual machines to run on a single physical server using a hypervisor. It improves resource utilization, reduces hardware costs, and enables flexible allocation of computing resources. Web services allow different applications to communicate over the Internet using APIs such as REST and SOAP. Together, virtualization and web services form the foundation of cloud computing by enabling efficient resource management, application integration, and scalable service delivery.

Analysis

Virtualization and web services are essential technologies in cloud computing. Virtualization provides scalability by allowing resources to be allocated dynamically based on demand, while web services enable seamless communication between cloud applications through standardized APIs. Their combined use improves performance, flexibility, reliability, and cost efficiency, making cloud platforms capable of delivering services to users anytime and from anywhere.

QUESTION 5



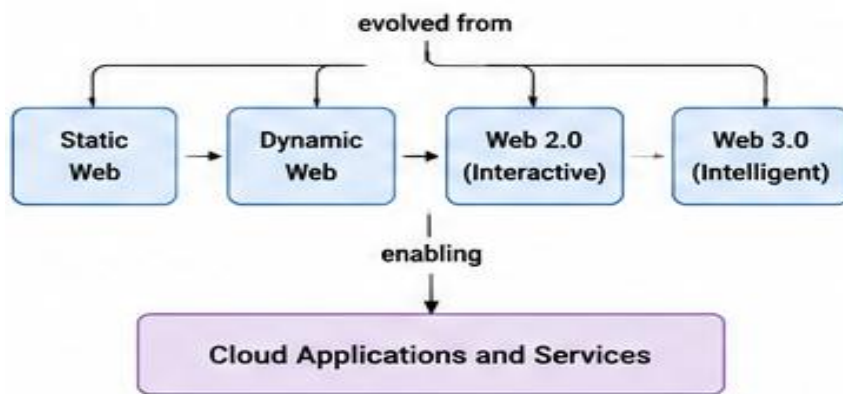
Explanation

Developing the concept map helped in understanding the overall architecture of cloud computing by visually connecting different concepts such as cloud services, virtualization, web services, deployment models, and scalability. The process made it easier to identify the relationships between these components and understand how they work together to provide reliable, flexible, and efficient cloud solutions. Creating the concept map also improved knowledge of cloud technologies by organizing complex information into a clear and structured format.

Analysis

The concept map demonstrates that cloud computing is an integrated system where multiple technologies work together to deliver services efficiently. Visualizing the relationships between concepts improved the understanding of cloud architecture and highlighted the importance of scalability, elasticity, virtualization, and web services in modern cloud environments. If the concept map were redesigned, it could include more real-world cloud platforms such as AWS, Microsoft Azure, and Google Cloud, along with additional security and monitoring services to provide a more comprehensive representation of cloud computing principles.

QUESTION 6



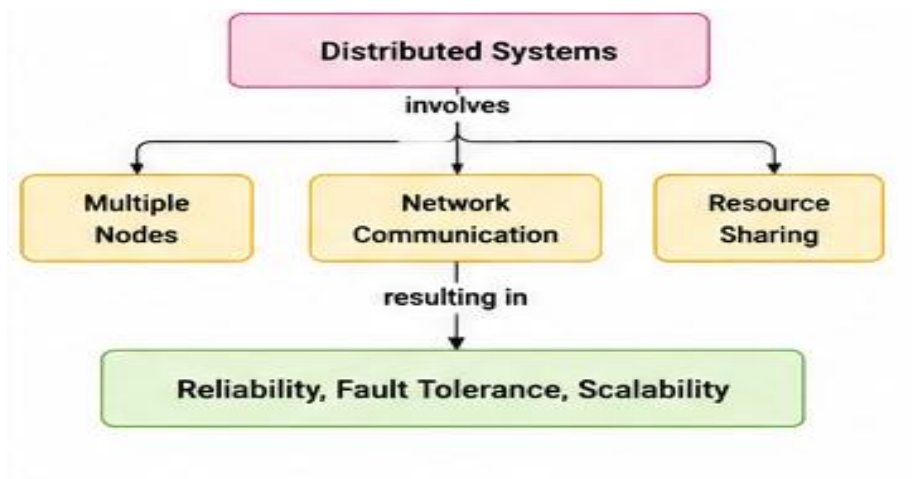
Explanation

Cloud deployment models define how cloud infrastructure is made available to users. The **Public Cloud** is owned and managed by a third-party provider and is accessible over the Internet. The **Private Cloud** is dedicated to a single organization, offering greater security and control. The **Hybrid Cloud** combines public and private clouds, allowing data and applications to move between them. The **Community Cloud** is shared by multiple organizations with similar security, compliance, or operational requirements.

Analysis

Each deployment model serves different business needs. Public clouds are cost-effective and highly scalable, making them suitable for startups and small businesses. Private clouds provide enhanced security and are ideal for organizations handling sensitive data. Hybrid clouds offer flexibility by combining the advantages of both public and private environments, while community clouds enable organizations with common objectives to share infrastructure securely. Choosing the right deployment model depends on factors such as cost, security, scalability, and organizational requirements.

QUESTION 7



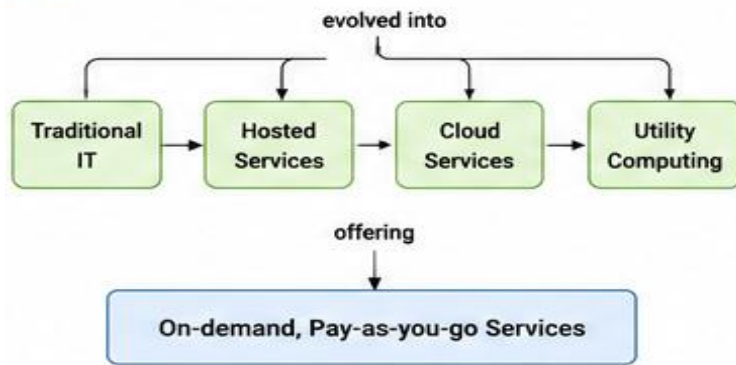
Explanation

Cloud actors are the participants involved in delivering and using cloud services. The **Cloud Provider** offers computing resources such as storage, servers, and applications. The **Cloud Consumer** uses these cloud services according to their requirements. A **Cloud Broker** acts as an intermediary between providers and consumers by managing service selection and integration. A **Cloud Carrier** provides the network connectivity that enables access to cloud services, while a **Cloud Auditor** evaluates cloud services to ensure they meet security, performance, and compliance standards.

Analysis

Cloud actors work together to ensure efficient cloud service delivery. The provider manages the cloud infrastructure, while the consumer accesses and uses the services. Brokers simplify service management, carriers ensure reliable communication, and auditors maintain trust through security and compliance assessments. Understanding the roles of cloud actors helps organizations design secure, reliable, and well-managed cloud environments.

QUESTION 8



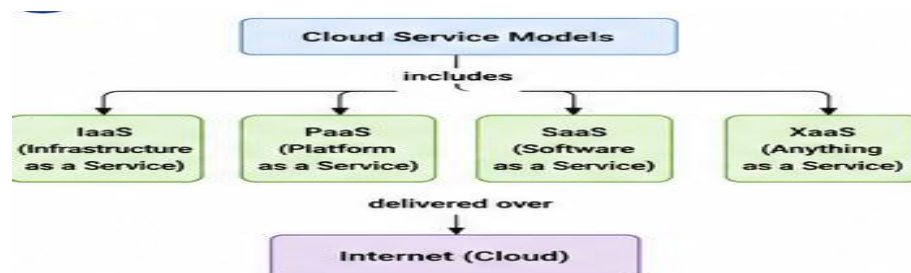
Explanation

Virtualization is the technology that allows multiple virtual machines (VMs) to run on a single physical server. It uses a **hypervisor** to create and manage virtual machines, enabling efficient sharing of hardware resources. Each virtual machine operates independently with its own operating system and applications. Virtualization improves resource utilization, reduces hardware costs, simplifies system management, and supports cloud computing by providing flexible and scalable infrastructure.

Analysis

Virtualization is one of the core technologies behind cloud computing because it enables efficient use of physical resources while reducing operational costs. It allows cloud providers to allocate computing resources dynamically based on user demand, improving scalability and flexibility. By isolating virtual machines, virtualization also enhances security, reliability, and system availability, making cloud environments more efficient and easier to manage.

QUESTION 9



Explanation

Web services enable communication and data exchange between different applications over the Internet. They use standard protocols and APIs to allow systems developed on different platforms to interact seamlessly. **REST (Representational State Transfer)** is a lightweight web service architecture that uses HTTP methods such as GET, POST, PUT, and DELETE. **SOAP (Simple Object Access Protocol)** is a protocol that exchanges structured XML messages and provides enhanced security and reliability. **APIs (Application Programming Interfaces)** act as interfaces that allow applications to access cloud services and exchange information efficiently.

Analysis

Web services are essential components of cloud computing because they enable interoperability between distributed applications. REST APIs are widely used due to their simplicity, speed, and scalability, making them suitable for modern web and mobile applications. SOAP is preferred in enterprise environments where high security and reliable messaging are required. Together, REST, SOAP, and APIs support seamless integration, efficient communication, and flexible service delivery in cloud-based systems.

QUESTION 10



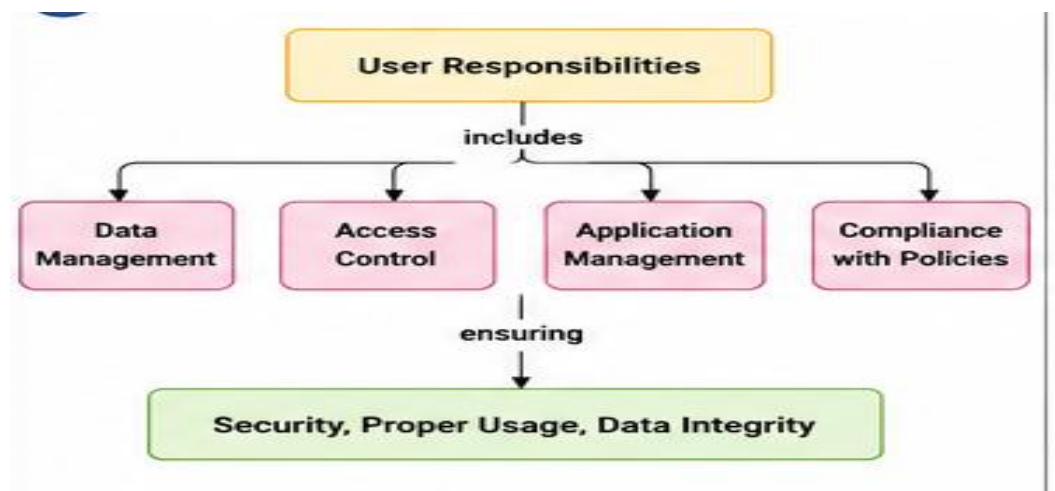
Explanation

Cloud computing is built on several essential characteristics that make it efficient and flexible. **On-demand self-service** allows users to access computing resources whenever needed without human intervention. **Scalability** enables resources to be increased or decreased according to workload. **Elasticity** automatically adjusts resources based on demand. **Resource pooling** allows multiple users to share computing resources securely and efficiently. **Measured service** monitors resource usage so that users pay only for what they consume.

Analysis

These characteristics make cloud computing highly efficient and cost-effective. Scalability and elasticity ensure that applications can handle changing workloads without performance issues. Resource pooling maximizes hardware utilization, while measured service supports the pay-as-you-use model, reducing unnecessary expenses. Together, these features improve system performance, availability, and flexibility, making cloud computing suitable for organizations of all sizes.

QUESTION 11



Explanation

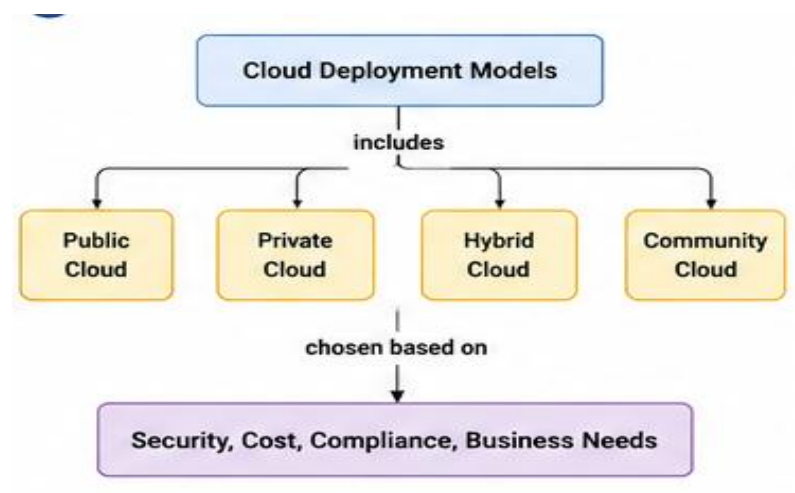
Scalability is the ability of a cloud system to increase or decrease computing resources based on workload requirements. It ensures that applications continue to perform efficiently as the number of users or data increases. There are two main types of scalability: **Vertical Scalability**

(Scale Up), which increases the capacity of a single server by adding more CPU, RAM, or storage, and **Horizontal Scalability (Scale Out)**, which adds more servers to distribute the workload. Scalability helps cloud applications maintain high performance and availability.

Analysis

Scalability is a key feature of cloud computing because it allows applications to adapt to changing user demands without affecting performance. Vertical scaling is simple but has hardware limitations, while horizontal scaling provides better fault tolerance and supports large-scale applications. By using scalable cloud infrastructure, organizations can improve system reliability, reduce operational costs, and deliver a better user experience during peak workloads.

QUESTION 12



Explanation

Elasticity is the ability of a cloud system to automatically allocate or release computing resources based on current demand. When the workload increases, additional resources are added automatically, and when the workload decreases, unnecessary resources are removed. This dynamic allocation ensures efficient resource utilization, reduces operational costs, and maintains application performance without manual intervention.

Analysis

Elasticity is one of the most important features of cloud computing because it enables systems to respond instantly to changing workloads. Unlike scalability, which focuses on increasing or decreasing capacity, elasticity emphasizes automatic and real-time resource adjustment. This helps organizations optimize costs, improve application availability, and maintain consistent performance during both high and low demand periods. Elastic cloud systems are especially beneficial for applications with unpredictable or seasonal traffic, ensuring efficient use of computing resources.

CONCEPT MAPPING

RUBRICS FOR CALCULATION

CONCEPT MAPPING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand